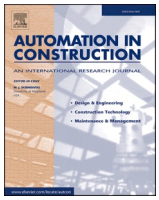
Automation in Construction 174 (2025) 106147



Contents lists available at ScienceDirect

Automation in Construction

journal homepage: www.elsevier.com/locate/autcon





Digital twin-enabled safety monitoring system for seamless worker-robot collaboration in construction

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ARTICLE INFO

Keywords:
Worker-robot collaboration
Digital twin
Safety monitoring
Construction robot

ABSTRACT

Worker-robot collaboration (WRC) has emerged as a transformative approach to augmenting the productivity of the construction industry. However, the development of a safety monitoring method or system for stopping robot operations in emergency is imperative, especially for seamless WRC on site. This paper presents a digital twin-enabled safety monitoring system for seamless WRC on site, characterized by its comprehensive perception of dynamic entities and dynamic calculation of protective separation distances during seamless WRC. The effectiveness of the proposed system is substantiated through a series of experiments, the result demonstrates its proficiency in mitigating collisions during robot operation in both static and dynamic WRC scenarios. The system achieves an average monitoring rate of 9.8 frames per second, an average reaction latency of 0.177 s, and a positional perception error of 0.09 m. It not only provides a practical tool for the implementation of seamless WRC on site, but also offers valuable insights for future WRC research.

1. Introduction

The construction industry has shown a keen interest in robotic technology to address issues such as labour shortage, low productivity, and poor working condition. Various automated construction robot systems have been developed [1], however, challenges arise when these systems are deployed in a highly dynamic and complex construction environment, leading to reduced precision and efficiency. Worker-robot collaboration (WRC) has been recognized as a promising method by leveraging the complementary skills of humans and machines to enhance the on-site usability of construction robots [2].

The predominant form of WRC in construction currently relies on remote control methods [3]. However, closer collaboration, both spatially and temporally, can transform the role of human workers from robot operators to teammates, enhancing the productivity and flexibility of on-site WRC [4]. Despite these benefits, safety concerns remain a significant barrier to seamless WRC in construction [5]. Since workers and construction robots share a same operational space in seamless WRC, physical risks arise due to planned or unexpected contact caused by the movement of either party [6]. When workers need to collaborate closely with robots to complete tasks, physical risks in WRC can further increase [7]. Therefore, an effective safety monitoring system is essential for achieving seamless WRC in construction.

Despite its critical importance, implementing a safety monitoring system on construction sites presents significant challenges. The unstructured and dynamic nature of construction scenarios may decrease the efficiency and precision of the safety monitoring system [8]. Traditional safety monitoring methods in industrial WRC, such as using physical or virtual fences to spatially separate robots and workers, are impractical. These methods tend to fragment a continuous workspace, reduce operational efficiency, and fail to adapt to rapidly changing construction environment. Therefore, there is a need for an advanced safety monitoring system with fine-grained monitoring capabilities and spatial flexibility that can be easily integrated into construction robots to support safe and seamless WRC on site.

This research proposes a digital twin-enabled safety monitoring system (SMS) for seamless WRC on construction sites. The contributions of this research are two-fold. First, it introduces a digital twin (DT)-based system capable of comprehensively perceiving and integrating the spatial information of workers, construction robots and construction materials during WRC, and establishing their approximate geometric occupation representations for spatial collision monitoring. Second, it proposes a rapid separation distance (SD) and a dynamic protective separation distance (PSD) calculation method for real-time safety monitoring in seamless WRC. The rest of this paper is organized as follows. Section 2 reviews related research on WRC in construction and

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https://doi.org/10.1016/j.autcon.2025.106147
Received 26 September 2024; Received in revised form 16 March 2025; Accepted 20 March 2025
Available online 25 March 2025
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safety monitoring methods for WRC. Section 3 introduces the research methodology. Section 4 details the proposed SMS. Section 5 presents the experiments to validate the system. Section 6 makes a conclusion and clarifies the limitations of this research.

2. Literature review

2.1. WRC in construction

Combining the hierarchical structure of construction works proposed by Bernold [9], with the Level of Robotic Automation (LoRA) defined by Beer et al. [10], it establishes a robust theoretical foundation for categorizing WRC in construction. Based on the work hierarchy at which WRC occurs, the collaborative system can be classified into three types: Division Collaboration (DC), Sequence Collaboration (SC), and Task Collaboration (TC) (Table 1). Each collaboration approach involves construction robots with different activity capabilities and corresponding LoRA that align with their designated collaboration roles.

DC focuses on appropriate labour division between workers and robots, such as task allocation [11,12] and scheduling planning [13,14]. Construction robots in DC are usually assumed to have a high LoRA (Levels 7–10) and can autonomously execute assigned tasks consisting of complex processes similar to workers. Through adequate labour division, workers and robots perform tasks independently in terms of time or space, resulting in clear temporal and spatial boundaries regarding their operation space [8]. Correspondingly, safety risks are minimized because there is minimal shared workspace between humans and robots, as well as physical risks.

SC involves a series of subtasks executed in a predetermined process to achieve the collaboration objective, with each subtask requiring specific skills from either workers or robots [15]. In SC, human workers typically handle the more fragile and dynamic subtasks, while robots undertake those with high physical demands. Related research focuses on designing optimal processes to maximize the full potential of human-robot collaboration. For example, MitterBerger et al. [2] developed a wood framework construction method, where robots hold and position wooden beams and workers then tie them together with ropes. Construction activities involving the installation and delivery of construction materials are a key focus of SC research [16]. For instance, Wu et al. [17] analyzed the efficiency of bricklaying teams, where robots lay bricks supplied by workers. In SC, robots' actions may need to be planned or authorized by human workers [18], but be executed autonomously by robots, demonstrating a middle LoRA in Levels 4–6. Nevertheless, the autonomy of robots and complex spatial interaction increases physical risks in WRC.

In TC, human workers and construction robots jointly complete a single challenging task, which involves the highest level of human-robot interaction (HRI). The task is typically characterized by the high demand of both physical and cognitive efforts, exceeding the sense and plan capabilities of current construction robots as well as the physical ability of human workers. Therefore, human workers usually act as operators and control construction robots manually or remotely through HRI devices, such as augmented reality (AR), virtual reality (VR) and force interaction devices [19]. For example, Chen et al. [20] presented a WRC approach for the sorting of construction wastes, where workers use AR devices to guide a robot based on their judgement of potential waste materials. Given the low LoRA (Levels 1–3) of construction robots in TC, physical risks are inherently controlled as robots are under human supervision.

In practical applications, WRC is usually based on hybrid configurations. For instance, a robot only possessing motion capability might initially collaborate with a worker to form a human-robot team for TC, thus engaging with another worker in process-level SC. Such hybrid WRC strategies become particularly prevalent when construction robots' LoRA remains insufficient to support their independent participation in higher-level collaborative activities. The operational compromise enables robotic systems to contribute to advanced construction operations while compensating for current technological limitations. However, the hybridity inevitably introduces systemic safety concerns across robotic applications - irrespective of their designated roles in WRC or attained LoRA levels. This further underscores the critical need for developing adaptive safety monitoring systems in WRC.

Table 1
WRC category based on construction works structure.

Collaboration approach	Collaboration level	Robot activity level	Required LoRA
Division Collaboration (DC)	Activity (e.g., foundation construction)	Process (e.g., welding)	10: Full Autonomy 9: Supervisory Control 8: Executive Control 7: Shared Control with Robot Initiative
Sequence Collaboration (SC)	Process (e.g., welding)	Work task (e.g., placing anchor)	6: Shared Control with Human Initiative 5: Decision Support 4: Batch Processing
Task Collaboration (TC)	Work task (e.g., placing anchor)	Motion/ move (e.g., grabbing)	3: Assisted Teleoperation 2: Tele-Operation 1: Manual

2.2. Safety monitoring method for WRC

Safety monitoring methods are pivotal in enhancing robotic safety [16]. Unlike industrial robots that are typically mounted on static bases, construction robots are usually mobile, comprising mobile platforms, kinematic structure, task-specific end-effectors, and integrated sensors [1]. The foundational architecture, first established by Shimizu Site Robot [21], remains fundamentally retained by modern construction robots. While dedicated safety standards for construction robots remain under development, ANSI/RIA R15.08 [22] provides critical normative guidance as the first safety standard for industrial mobile robots (IMR). Particularly, its classification of Type C IMR, defined as "mobile platforms integrated with manipulator attachments", is a configuration structurally analogous to modern construction robots. The standard mandates that manipulators mounted on mobile platforms comply with ISO 10218 [23] requirements, which specifies four principal collaborative methodologies: Safety-Related Monitoring Stop (SRMS), Hand Guidance (HG), Speed and Separation Monitoring (SSM), and Power and Force Limiting (PFL). The subsequent publication of ISO 15066 [24] significantly advances WRC safety frameworks by providing detailed design guidelines for safe WRC methods. Therefore, different safety monitoring methods can be designed based on WRC methods adapted in production.

SRMS demands that robotic operations are conducted at a predetermined distance from human workers, commonly achieved through the implementation of physical or virtual barriers to demarcate zones separating humans and robots. While effective and widely used in industrial environments, the method has been criticized for creating spatial segmentation and may potentially undermine on-site productivity [8]. The HG method facilitates teaching and programming robots through direct physical interaction [25], however, SRMS must be the prerequisite for HG. SSM defines separation distance (SD) and protective separation distance (PSD) based on the speed of humans and robots, dictating that robotic actions are permissible when SD exceeds PSD. Since SSM requires simultaneous detection of the spatial position and speed of both humans and robots, the corresponding safety monitoring method requires higher sensing resources. PFL safeguards human workers from physical harm by limiting the force output of robots during contact [26]. However, PFL serves as a post-contact protective measure and is not applicable when robots possess surface dangers, such as

holding drills or welding torches in their end effectors. Therefore, SSM stands out as the most appropriate and adaptable method for ensuring safe WRC on site. With the development of sensing technologies, SSM is becoming increasingly cost-effective and feasible.

The key of SSM is to calculate SD and PSD according to the spatial motion information of both robots and collaborating workers, which requires advanced robot intelligence. For monitoring human bodies within collaborative workspace, sensors with three-dimensional perception capabilities, such as light detection and ranging (LiDAR) [27] and depth cameras [28], are commonly employed. For seamless WRC, limb-level monitoring becomes imperative for accurate SD sensing. The effectiveness of SD sensing is heavily influenced by how obstacles are represented. For instance, Lucci et al. [26] utilized depth cameras for the limb-level detection of human workers, but the collision representation relied on point clouds, which are less versatile than surface-based representations like triangle meshes. Similarly, Rosenstrauch et al. [29] used Kinect V2 for human body detection, with a simplified representation consisting of joint keypoints without collision volumes. Li et al. [30] also adopted joint keypoints for SD calculations. However, in close WRC scenarios, keypoints may not always represent the closest points on a worker's body relative to a robot, potentially leading to inaccurate estimation for SD. Surface-based representations for both workers and robots can provide more accurate SD measurements. Nevertheless, overly complex representations, such as the Skinned Multi-Person Linear (SMPL) model [25], need substantial computing resources, thus hindering real-time performance. Simplified geometric representations, including spheres, boxes, and capsules [31,32], offer a viable alternative, but the level of detail (LoD) must be carefully designed to strike a balance between computational efficiency and representation accuracy.

While ISO 15066 provides calculation methods for PSD, obtaining necessary value for calculation, particularly the speed of workers and robots, renders it challenging. A common approach is to set a constant value for workers' speed to estimate PSD [33–35], where a maximum speed assumption for workers is often established for safety. However, the assumption also constrains robots' workspace, potentially reducing the efficiency and flexibility of WRC [36]. Besides, current research for safety monitoring concentrates on WRC in manufacturing, where robots usually operate on a tabletop scale, using stationary arms to handle small objects. However, in construction sites, components and objects such as drywall panels manipulated by robots may have a large size and should not be overlooked [18]. In summary, although SSM is the most suitable collaboration method for WRC in construction, there is still a lack of an appropriate safety monitoring system with capabilities of 1) balancing the computational accuracy and efficiency of SD, 2) dynamically calculating PSD, and 3) monitoring objects held by robots to support safe WRC in construction.

3. Methodology

This research employs a constructive research approach [37], encompassing three phases: problem identification, method development, and validation. The challenges identified in current safety monitoring methods or systems for WRC on site include a lack of efficient and dynamic sensing methods for SD and PSD, as well as integrating detection for manipulated materials by construction robots during monitoring. To address the gap, this research introduces a digital twin-enabled SMS for WRC. The conceptual framework of the proposed system is depicted in Fig. 1.

Digital twin fundamentally comprises three core components: physical assets in operational environment, their synchronized virtual replica in cyberspace, and bidirectional data pipelines enabling data and information exchange between the two domains [38]. Through the integration of multimodal sensors, heterogeneous data and computational intelligence, digital twin establishes a bridge between physical and virtual environments. The intrinsic demand for DT-enabled safety monitoring stems from the inherent complexity of construction robots. Beyond safety monitoring, construction robots require basic capabilities in path planning and motion control to execute tasks. Maintaining functional decoupling while achieving resource reuse across subsystems becomes critical in such complex architectures. In the proposed system, safety monitoring functions as an upper-layer application are built upon the foundational digital twin infrastructure, thereby achieving systemic decoupling from other operational functions. It is crucial that the virtual representation of physical assets, as a digital avatar, can be simultaneously leveraged for diverse operational requirements - from real-time path planning to high-level situational awareness - beyond its primary safety monitoring purpose.

In the SMS, the perception of physical assets, including robots, workers, and materials, is achieved through sensors integrated with robots. The sensors collect environmental data, from which algorithms extract the spatial poses, geometric representations, and velocities of physical assets. The information is then represented as virtual assets or values within the digital twin system. The virtual assets or values provide essential information for calculating SD and PSD according to the SSM method outlined in ISO 15066 [24]. SD and PSD are compared circularly to assess the real-time safety of WRC. If PSD is exceeded by SD, the system accesses the physical actuators of robots to stop it, thereby

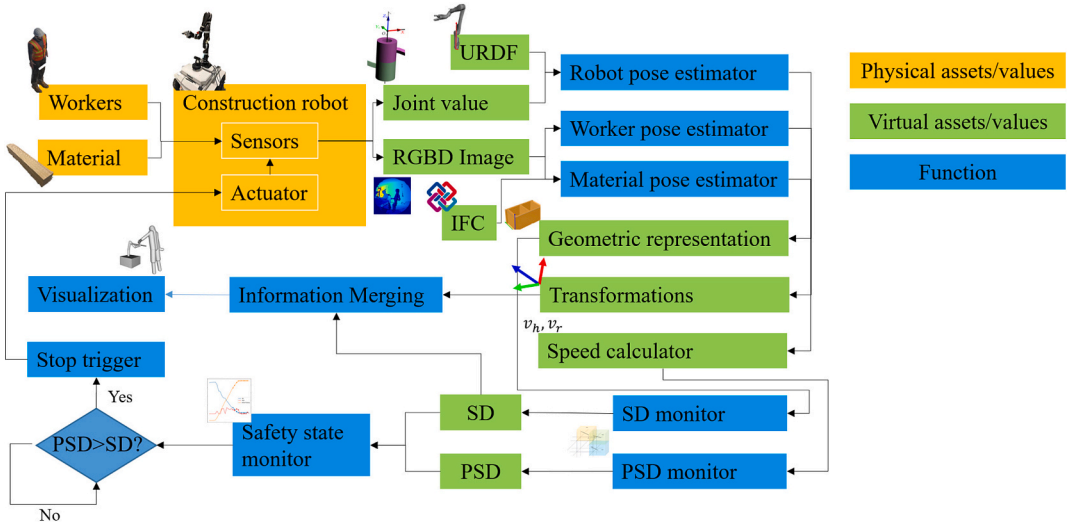


Fig. 1. Framework of the proposed SMS.

safeguarding workers from potential physical hazards.

4. Safety monitoring method

The proposed SMS mainly comprises two components: the perception of physical assets, which provides basic information for detecting safety states, and safety monitoring based on SSM, which offers the criteria for assessing safety states.

4.1. Perception of physical assets in the digital twin system

Physical assets for WRC in construction include construction robots, construction workers, and construction materials. Due to the heterogeneity of their movement patterns and physical characteristics, it is necessary to adopt different approaches to perceive their spatial poses and occupancy so as to construct their virtual representations in the digital twin system.

4.1.1. Construction robot

The construction robot prototype used in this research (Fig. 2a) consists of 1) a four-wheel omnidirectional mobile chassis, 2) a 7-degree-of-freedom (DoF) Kinova Jaco 2 robotic arm with a maximum payload of 2.6 kg, 3) a two-finger gripper as the end-effector, and 4) an attached depth camera for environmental perception. The configuration of this robot mirrors that of the previously described general construction robot configuration. As a prototype, it can perform general WRC tasks including material transportation (e.g., handling materials for workers) and cooperative operations (e.g., workpiece stabilization during work machining tasks).

To establish a digital twin representation of construction robots in virtual environments, it needs to sense the spatial poses of their moving components (represented via coordinate systems) and model their spatial occupancy. The mobile base of construction robots is the primary reference as the base coordinate system. The X-axis of the base coordinate system is aligned with the positive direction of robots' movement, while the *Z*-axis is aligned with the upward vertical direction and the Y-axis is determined by right-hand rules (Fig. 2a). The spatial occupancy representation of mobile bases in the virtual environment is based on a simplified cuboid model that approximates their physical dimensions.

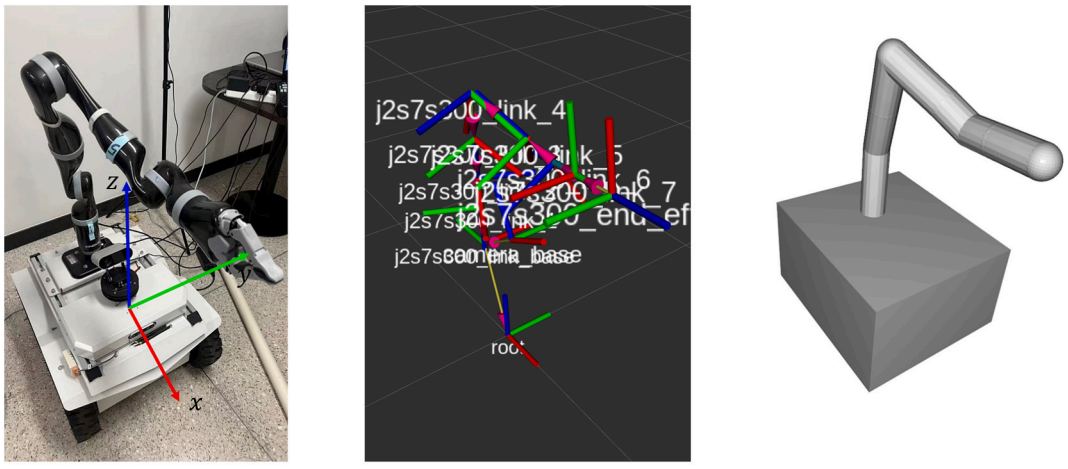
The kinematic structure of construction robots, such as robotic arms, can be modelled as the assembly of multiple rigid bodies interconnected through rotational and translational joints. The pose of kinematic structure can be comprehensively represented using individual coordinate systems of its joints. The Denavit-Hartenberg (DH) method provides a parameterized approach to describe the transformations among the coordinate systems using four parameters, including α, d, a, and θ. Taking the 7-DoF robotic arm adopted by the prototype as an example, α, d, and a are constants related to the inherent dimensions of the links. The joint angles θ of its rotational joints are the sole variables and can be measured through the integration of rotary encoders within the joints. The Unified Robot Description Format (URDF) is utilized to describe and store the constant DH parameters of construction robots, complemented by the real-time sensing of joint angles to reconstruct the pose of kinematic structure. Given that the links of robotic arm are generally cylindrical in shape, it is suggested to use capsules to represent their spatial occupancy in the virtual environment (Fig. 2c).

Vision sensors are critical for construction robots to collect external information in a dynamic environment. Accurately determining the pose of sensors within the base coordinate system is crucial for robots to precisely perceive the pose of other external objects in the environment. Eye-in-hand calibration is utilized to obtain the precise pose of sensors relative to the base coordinate system, denoted as T_c^0. However, the geometric representation of sensors are commonly unnecessary, as they are typically fixed to mobile bases and remain stationary. Therefore, its spatial occupancy is generally integrated into mobile bases. The perceiving and representing results of construction robots are shown in Fig. 2.

4.1.2. Human worker

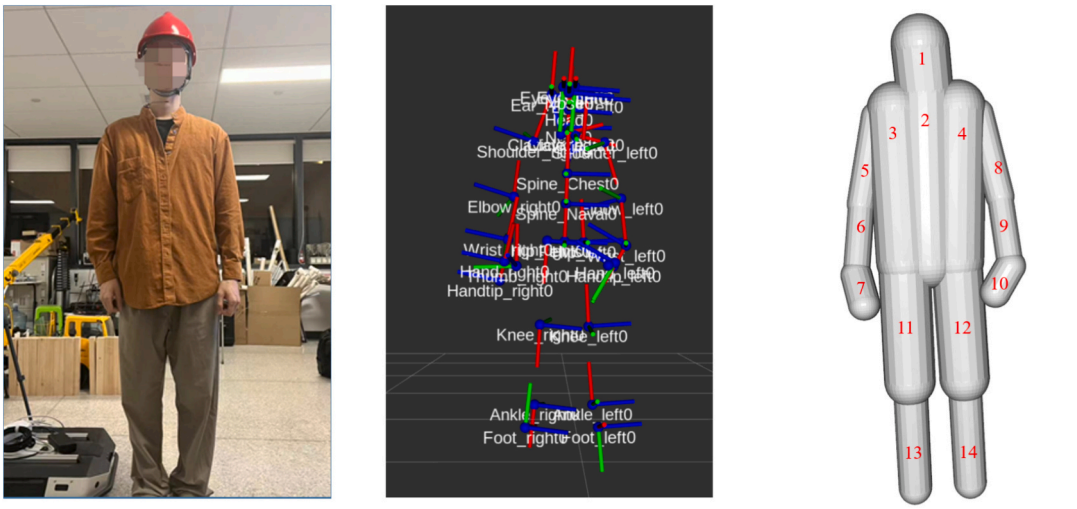
To accurately perceive workers' poses, this research employs a 3D human pose estimation algorithm to derive a skeletal model from RGB-depth (RGBD) images captured by depth cameras. Commonly-employed skeletal models for human pose estimation include the 17-point model delineated by Common Objects in Context (COCO) [39] and the 32-point model defined by Azure Kinect body tracking (AKBT) [40]. The AKBT software development kit is utilized for pose estimation, which has demonstrated superior real-time performance and accuracy in human factors and safety applications [41]. The pose estimation process involves initially segmenting human masks from images using a neural network and estimating the position of individual's joints. Then, depth values are assigned to each pixel of human masks based on depth data, and each pose of joint is determined via random forest voting. As a result, AKBT outputs a 32-point skeletal model, as depicted in Fig. 3b. Denoting the pose of a particular joint to the depth camera sensor coordinate system as T_j^c, then its relative pose to the base coordinate system can be calculated as $T_j^0 = T_c^0 T_j^c$.

A skeletal model serves as a reference for the virtual representation



a) Physical construction robot b) Pose estimation c) Virtual representation

Fig. 2. Perceiving and representing results of construction robots.



a) Physical human worker b) Pose estimation c) Virtual representation

Fig. 3. Perceiving and representing results of human workers.

of a human body. Capsules are ideal geometric representations due to their resemblance to the shape of human limbs with minimal parameters. Given a limb with its two-joint positions $j_1 = [x_1, y_1, z_1]$ and $j_2 = [x_2, y_2, z_2]$, the capsule representation of its spatial occupancy can be defined by seven parameters (Eq. (1)).

$$capsule = \{x_1, y_1, z_1, x_2, y_2, z_2, r\} \tag{1}$$

where, r represents the radius of capsule.

The radius values are defined according to the standard "GB/T 10000-2023 Human dimensions of Chinese adults" [42] based on the 50th percentile of static anthropometric dimensions for adult males aged 18–70 in various body parts. For instance, the 50th percentile head circumference is 570 mm, thus the radius r_1 of capsule representing head occupancy is $570/2\pi$ approximately 91 mm. For the palm, since its cross-section is not approximately circular, the palm width (100 mm) is used as the reference for calculating radius r_7, that is, 50 mm. According to the simplified skeletal model, a worker's spatial occupancy representation is composed of 14 capsules (Fig. 3c), with each capsule's radius detailed in Table 2.

4.1.3. Material

Due to the diverse forms of construction materials, it is essential to establish a consistent standard to define their coordinate systems to unambiguously describe their poses. According to their approximate geometric forms, common materials can be categorized into volumetric, planar, and linear types. Their coordinate systems can be defined as follows.

Table 2
Spatial occupancy representation parameters for human body parts.

Body part (Fig. 3c)	Joints	Capsule Radius (mm)	Standard reference (mm)
Head (1)	Neck-Forehead	91	570 (Head circumference)
Upper arm (5,8)	Shoulder-Elbow	47	295 (Upper arm circumference)
Forearm (6, 9)	Elbow-Wrist	42.5	267 (Forearm circumference)
Hand (7, 10)	Wrist-Finger tip	50	100 (Palm width)
Torso (2, 3, 4)	Shoulder-Hip (3 , 4) Neck-Pelvis(2)	101.5	203 (Chest depth)
Thigh (11,12)	Hip-Knee	85.5	537 (Thigh circumference)
Calf (13, 14)	Knee-Ankle	57.5	361 (Calf circumference)

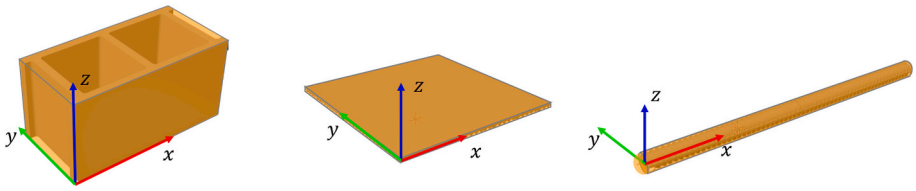
4.1.3.1. Volumetric material. Materials possessing comparable dimensions in three dimensional space, such as prefabricated beams, slabs, masonry blocks, and I-beams, are considered volumetric. The *Z*-axis of the volumetric material's coordinate system should be vertically upward and aligned with one of its edges, while the X-axis aligns with the longer side of the remaining two ones, and the Y-axis aligns with the shorter one (Fig. 4a).

4.1.3.2. Planar material. Materials with comparable dimensions in length and width but exhibiting minimal thickness, such as tiles, ceilings, curtain wall panels, doors, and windows, are classified as planar. Planar materials often exhibit surface anisotropy. For instance, tiles have distinct glazed and rough surfaces. The *Z*-axis of the planar material's coordinate system is defined to point toward the "external" side after installation if materials exhibit surface anisotropy (relative to the side contacting the working surface), while the X-axis aligns with the longer side of the remaining two ones, and the Y-axis aligns with the shorter one (Fig. 4b).

4.1.3.3. Linear material. Materials with significant length but relatively small dimensions in the other two dimensions, such as rebar, rods, and pipes, are considered linear. Linear materials are typically geometrically symmetrical about the central axis, such as scaffolding pipes, drainage pipes, and rebar, with circular cross-sections. For linear materials, the X-axis of their coordinate system should align with the longitudinal axis of materials. If linear materials are rotationally symmetric about the X-axis, the Y and Z axes can be arbitrarily oriented; otherwise, they should follow the same definitions as volumetric or planar materials (Fig. 4c).

Fig. 4 shows the geometric representation of materials in the virtual environment, designed to closely resemble their actual shapes. Volumetric and planar materials are typically represented by cuboids, while linear materials are represented by capsules.

Given the large size of many materials and the difficulty in fully perceiving their shapes, this research employs an ArUco tag-based method for the perception of material pose and virtual representation reconstruction. ArUco tags are square markers with a black border and a binary matrix. The black border facilitates tag detection in images, while the internal binary code is used for tag identification and error correction. ArUco tags are known for their rapid deployment and robust recognition, making them widely used in augmented reality and mixed



a) Volumetric material c) Planar material b) Linear material

Fig. 4. Coordinate system definition and geometric representation for materials.

reality applications requiring pose estimation. ArUco tags are attached to materials to be perceived (Fig. 5a). Given camera's intrinsic parameters and the size of ArUco markers, the pose of ArUco tag relative to the camera coordinate system can be estimated, denoted by a transformation matrix T_a^c. To determine the position of materials relative to camera sensors, the pose of the ArUco tags relative to the material coordinate system must also be obtained through pre-measurement, denoted as T_m^a. Then, the pose of materials related to the base coordinate system can be calculated as $T_m^0 = T_c^0 T_a^c T_m^a$.

To completely perceive the pose and shapes of materials, T_m^a and material shapes are stored and transmitted using the Industrial Foundation Classes (IFC) data. Since ArUco tags can only store limited information, this research combines quick response (QR) codes with ArUco tags for comprehensive perception, as shown in Fig. 5a. In the process of perceiving, ArUco tags are first detected to obtain T_a^c, simultaneously narrowing the detection range for QR codes. Reading QR codes yields the *GlobalID* of materials, consisting of a 22-digit alphanumeric sequence. Based on the *GlobalID*, the system searches the stored IFC data for corresponding *ifcRelDefinesByProperties* parameter, which includes the images of ArUco tags (*ifcLabel*) and T_m^a, to construct the pose of materials (Fig. 5b). Materials' shape information is parsed from *ifcShapeRepresentation*. The hierarchical structure of properties stored in IFC data, which can be easily constructed by IFC-supported modelling software, is shown in Fig. 6. In the end, the pose and representation of materials are reconstructed in the virtual environment (Fig. 5c).

4.2. Safety monitoring based on SSM

The key to SSM-based safety monitoring lies in comparing SD with PSD as a threshold. SD is an objective physical quantity, and its rapid calculation is crucial to meet the real-time requirements of safety monitoring. Meanwhile, the calculation method for PSD should be designed to account for dynamic collision risks in WRC.

4.2.1. Separation distance calculation

Based on the pose and geometric representations of physical assets perceived by the digital twin system, SD is expressed as the minimum distance between the geometric representations of a human worker and a construction robot (including both the robot and material it handles). To calculate SD, it must consider all possible configurations of workers' limbs relative to the components of robots. Human workers' representations are a capsule set $\mathscr{C}_h$, and robots' representations include a capsule set $\mathscr{C}_r$ and a box set $\mathscr{B}_r$. Given that all entities' representations are designed as the combinations of capsules or boxes, the calculation of SD needs to determine the spatial distances of 1) capsule-to-capsule, and 2) capsule-to-box.

4.2.1.1. Capsule-to-capsule distance. For two capsules denoted as $c_1 = \{x_1^1, y_1^1, z_1^1, x_1^2, y_1^2, z_1^2, r_1\}$ and $c_2 = \{x_2^1, y_2^1, z_2^1, x_2^2, y_2^2, z_2^2, r_2\}$, their distance s is defined by Eq. (2).

$$s(c_1, c_2) = d(\overline{P_1P_2}, \overline{Q_1Q_2}) - r_1 - r_2 \tag{2}$$

where, $P_1 = (x_1^1, y_1^1, z_1^1); P_2 = (x_1^2, y_1^2, z_1^2); Q_1 = (x_2^1, y_2^1, z_2^1); Q_2 = (x_2^2, y_2^2, z_2^2)$; $d(\overline{P_1P_2}, \overline{Q_1Q_2})$ is the shortest distance between $\overline{P_1P_2}$ and $\overline{Q_1Q_2}$.

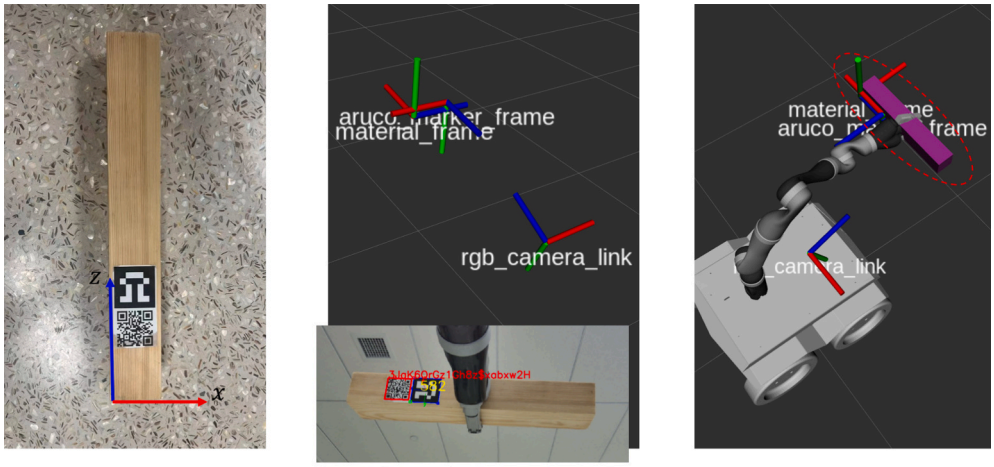
Let $\boldsymbol{s_1} = P_2 - P_1, \boldsymbol{s_2} = Q_2 - Q_1$, a shortest line segment $\overline{PQ}$ between $\overline{P_1P_2}$ and $\overline{Q_1Q_2}$ exists, the length of which is expressed by Eqs. (3)–(5).

$$P(\lambda_1) = P_1 + \lambda_1 \boldsymbol{s_1} \tag{3}$$

$$Q(\lambda_2) = Q_1 + \lambda_2 \boldsymbol{s_2} \tag{4}$$

$$d(\overline{P_1P_2}, \overline{Q_1Q_2}) = |\overline{PQ}| = \|(P_1 + \lambda_1 \boldsymbol{s_1}) - (Q_1 + \lambda_2 \boldsymbol{s_2})\| \tag{5}$$

To minimize $|\overline{PQ}|$, let $\frac{\partial \|\boldsymbol{PQ}\|}{\partial \lambda_1} = 0, \frac{\partial \|\boldsymbol{PQ}\|}{\partial \lambda_2} = 0$, thus:



a) Physical material b) Pose estimation c) Virtual representation

Fig. 5. Perceiving and representing results of materials.

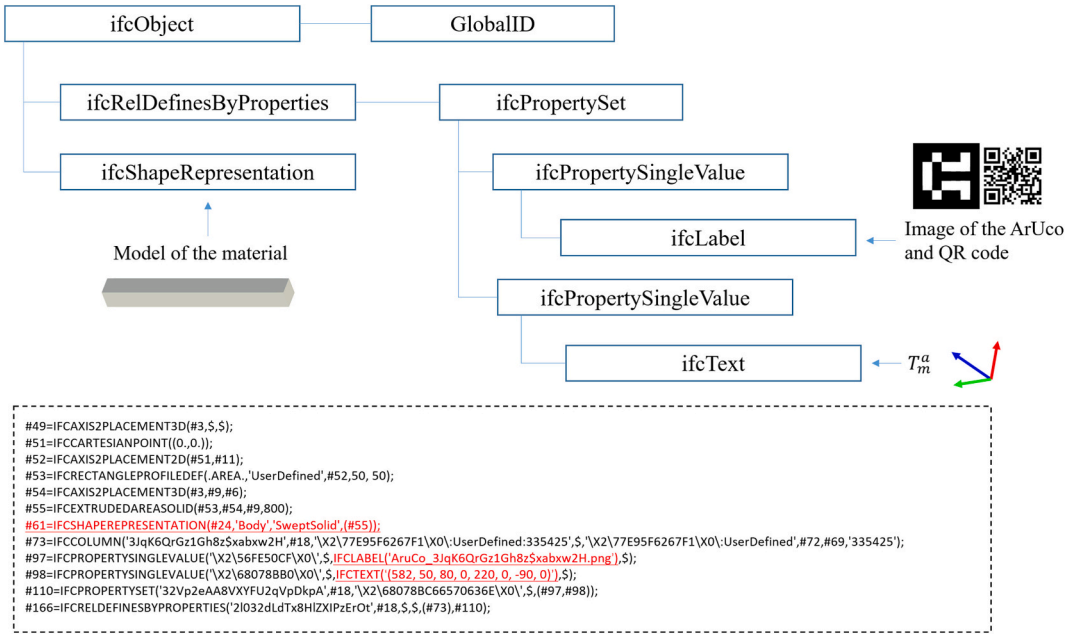


Fig. 6. Hierarchical structure of material properties stored in IFC data.

$$\begin{cases} \lambda_1 = \dfrac{(\boldsymbol{s_1}\cdot\boldsymbol{s_2})\left(\overrightarrow{P_1Q_1}\cdot\boldsymbol{s_2}\right) - \|\boldsymbol{s_2}\|^2\left(\overrightarrow{P_1Q_1}\cdot\boldsymbol{s_1}\right)}{\|\boldsymbol{s_1}\|^2\|\boldsymbol{s_2}\|^2 - (\boldsymbol{s_1}\cdot\boldsymbol{s_2})^2} \\ \lambda_2 = \dfrac{-(\boldsymbol{s_1}\cdot\boldsymbol{s_2})\left(\overrightarrow{P_1Q_1}\cdot\boldsymbol{s_1}\right) - \|\boldsymbol{s_1}\|^2\left(\overrightarrow{P_1Q_1}\cdot\boldsymbol{s_2}\right)}{\|\boldsymbol{s_1}\|^2\|\boldsymbol{s_2}\|^2 - (\boldsymbol{s_1}\cdot\boldsymbol{s_2})^2} \end{cases} \tag{6}$$

Values of λ_1, and λ_2 between 0 and 1 indicate that P lies on $\overline{P_1P_2}$ and Q lies on $\overline{Q_1Q_2}$, in which case, $d(\overline{P_1P_2}, \overline{Q_1Q_2})$ is calculated based on Eq. (5). Otherwise, the shortest distances from P_1, P_2 to $\overline{Q_1Q_2}$, and Q_1, Q_2 to $\overline{P_1P_2}$ must be calculated separately to determine $d(\overline{P_1P_2}, \overline{Q_1Q_2})$, as expressed by Eqs. (7)–(10) (exemplified by the distance from P_1 to $\overline{Q_1Q_2}$).

$$f(P_1, \overline{Q_1Q_2}) = \|P_1 - (Q_1 + g(\lambda_2)\, s_2)\| \tag{7}$$

$$\lambda_2 = \frac{(s_1\cdot s_2)[(P_1 - Q_1)\cdot s_2] - \|s_2\|^2[(P_1 - Q_1)\cdot s_1]}{\|s_1\|^2\|s_2\|^2 - (s_1\cdot s_2)^2} \tag{8}$$

$$g(\lambda) = \begin{cases} 0, \lambda < 0 \\ \lambda, 1 < \lambda < 0 \\ 1, \lambda > 1 \end{cases} \tag{9}$$

$$d(\overline{P_1P_2}, \overline{Q_1Q_2}) = min\{f(P_1, \overline{Q_1Q_2}), f(P_2, \overline{Q_1Q_2}), f(Q_1, \overline{P_1P_2}), f(Q_2, \overline{P_1P_2})\} \tag{10}$$

In the end, the capsule-to-capsule distance $s(c_1, c_2)$ is obtained by substituting Eq. (10) or Eq. (5) into Eq. (2).

4.2.1.2. Capsule-to-box distance. To rapidly calculate the minimum distance between a capsule and a box, the 3D space is divided into 27 Voronoi cells according to the six faces of the box, as shown in Fig. 7. Each vertex, edge, and face of the box defines a unique Voronoi domain. It can be proven that, for any point within a domain, the closest distance from the point to the box always lies on the domain's corresponding geometric element. For instance, if a point resides within a vertex domain, then the corresponding vertex is guaranteed to be the nearest point on the box to that point. This principle also applies to lines within the domain, which could greatly simplify the calculation.

For capsule $c = \{x^1, y^1, z^1, x^2, y^2, z^2, r\}$ and $P_1 = (x^1, y^1, z^1)$, $P_2 = (x^2, y^2, z^2)$, defining $v_i(\overline{P_1P_2})$ as the shortest distance from $\overline{P_1P_2}$ to the box within domain V_i and $\overline{P_1'P_2'}$ as the segmentation of $\overline{P_1P_2}$ in domain V_i, $v_i(\overline{P_1P_2})$ is calculated as follows.

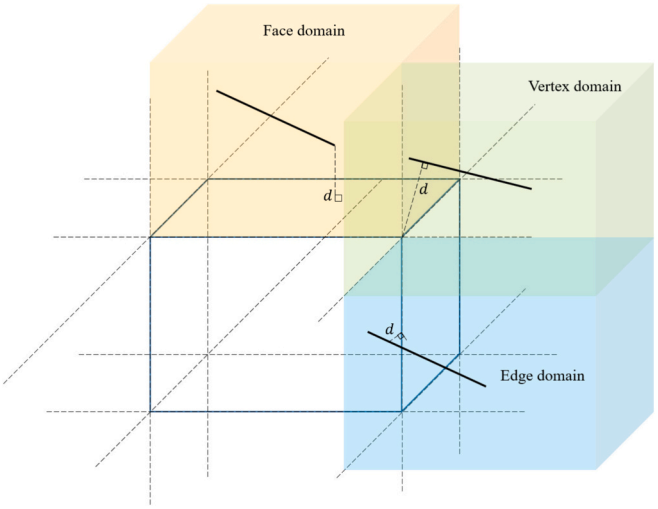


Fig. 7. Divided Voronoi regions for a spatial box, along with the illustration of the shortest distance from lines to the box within domains.

1) If $\boldsymbol{V_i}$ is a vertex domain, $\boldsymbol{v_i}(\overline{\boldsymbol{P_1P_2}})$ equals the shortest distance from the vertex to $\overline{\boldsymbol{P_1'P_2'}}$, which can be referred to Eqs. (7)–(10);
2) If $\boldsymbol{V_i}$ is an edge domain, $\boldsymbol{v_i}(\overline{\boldsymbol{P_1P_2}})$ equals the shortest distance from the edge to $\overline{\boldsymbol{P_1'P_2'}}$, which can be calculated using Eqs. (3)–(6);
3) If $\boldsymbol{V_i}$ is a face domain, $\boldsymbol{v_i}(\overline{\boldsymbol{P_1P_2}})$ equals the shortest distance from $\boldsymbol{P_1}'$ or $\boldsymbol{P_2}'$ to the face.

Finally, the capsule-to-box distance is calculated as Eq. (11). With the capsule-to-capsule and capsule-to-box distance calculating methods, SD can be calculated as Eq. (12).

$$s(c, b) = d(\overline{P_1P_2}, b) - r = min\{v_i(\overline{P_1P_2})\} - r \tag{11}$$

$$SD = min\left\{ \min_{c_h \in \mathscr{C}_h, c_r \in \mathscr{C}_r} (s(c_h, c_r)), \min_{c_h \in \mathscr{C}_h, b_r \in \mathscr{B}_r} (s(c_h, b_r)) \right\} \quad (12)$$

4.2.2. Protective separation distance calculation

The design of PSD should be universally applicable to all types of construction robots, irrespective of their size, shape, or functionality. Achieving the universality necessitates a focus on the intrinsic safety of human-robot collaboration. Fundamentally, the risk of human-robot collisions stems from kinetic energy possessed by construction robots and their components. The core principle of PSD is to halt robots before such collisions occur, or at the very least, to ensure that robots dissipate their kinetic energy upon impact. To apply the principle across various robotic platforms, PSD design must account for the differing stopping times required by different robots and the performance capabilities of their sensing systems. The ISO 15066 standard defines the general equation for calculating PSD as Eq. (13).

$$S_p(t_0) = S_h + S_r + S_s + C + Z_d + Z_r \quad (13)$$

where, $S_p(t_0)$ is PSD at the time t_0; C is the intrusion distance that a part of human body can intrude into the sensing field before it is detected (ISO 13855 [43]); Z_d and Z_r are the position uncertainty of human workers and robots respectively, caused by perception errors. Eqs. (14)–(16) further express S_h, S_r, and S_s, respectively.

$$S_h = \int_{t_0}^{t_0+T_r+T_s} v_h(t)dt \quad (14)$$

$$S_r = \int_{t_0}^{t_0+T_r} v_r(t)dt \quad (15)$$

$$S_s = \int_{t_0+T_r}^{t_0+T_r+T_s} v_s(t)dt \quad (16)$$

where, T_r is the reaction time of robots, including time for perception, processing and activation of a robot stop triggering; T_s is the stopping time of robots, including time for stop signal transmission and braking; v_h is the approaching speed of workers relative to robots; v_r is the approaching speed of robots relative to workers during reaction; v_s is the approaching speed of robots relative to workers during stopping.

By combining Eqs. (13)–(16), Eq. (17) is formed to indicate that PSD consists of the relative distance decreased between workers and robots during reaction time and stopping time, plus constant term. Triggering stop when SD exceeds PSD ensures that robots will stop running before any possible collision happen between workers and robots (Fig. 8), thereby guarding safety in WRC.

$$S_p(t_0) = \int_{t_0}^{t_0+T_r} (v_h(t) + v_r(t))dt + \int_{t_0+T_r}^{t_0+T_r+T_s} (v_h(t) + v_s(t))dt + C + Z_d + Z_r \quad (17)$$

To estimate dynamic PSD with sufficient margin for safety, certain approximations and simplifications need to be applied. This research makes three assumptions: 1) workers maintain a constant approaching speed after t_0; 2) construction robots maintain a constant approaching speed during T_r, and decrease their speed linearly to zero during T_s; and 3) the speed directions of workers and robots are along their shortest distance. According to these assumptions, Eq. (17) is simplified as Eq. (18).

$$S_p(t_0) = v_h(t_0)(T_r + T_s) + v_r(t_0)T_r + \frac{1}{2}v_r^2(t_0)T_s + C + Z_d + Z_r \quad (18)$$

Further, according to the current and historical poses of workers and robots, $v_h(t_0)$ and $v_r(t_0)$ can be calculated as Eqs. (19)–(20).

$$v_h(t_0) = \frac{\Delta S_h(t_0)}{\Delta t} = \frac{S(p_h(t_0), p_r(t_0 - \Delta t)) - S(p_h(t_0), p_r(t_0))}{\Delta t} \quad (19)$$

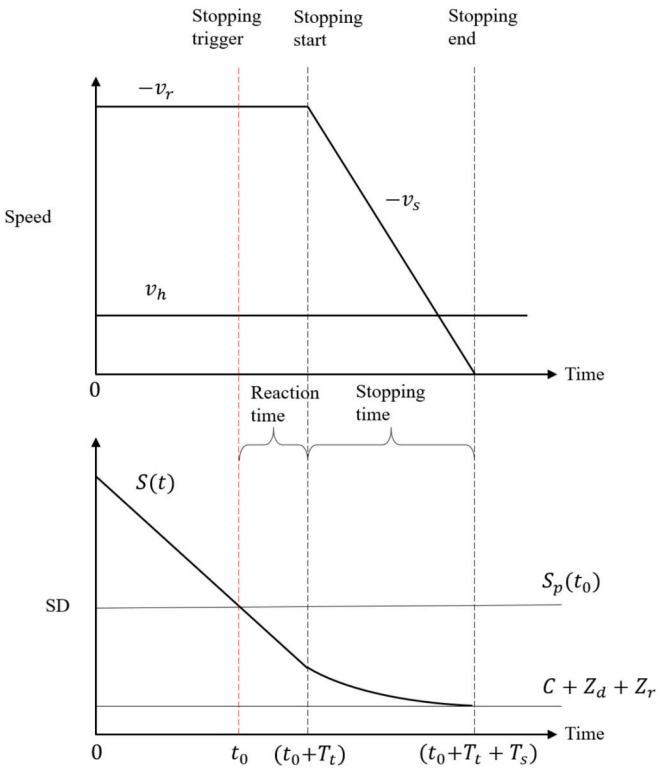


Fig. 8. Relationship between SD and PSD over time.

$$v_r(t_0) = \frac{\Delta S_r(t_0)}{\Delta t} = \frac{S(p_h(t_0 - \Delta t), p_r(t_0)) - S(p_h(t_0), p_r(t_0))}{\Delta t} \quad (20)$$

where, $p_h(t_0)$ is the position of worker's body part at t_0; $p_r(t_0)$ is the position of robot at t_0; Δt is the sampling intervals of SMS for worker's and robot's poses.

Substituting Eqs. (19)–(20) into Eq. (18) enables the dynamic calculation of PSD at t_0. Ultimately, a safety state monitor is deployed to monitor current SD and dynamic PSD in WRC, so as to control the actuator of construction robots to stop when SD is less than PSD.

5. Experiment and result

To evaluate the feasibility and performance of the proposed SMS, a series of experiments were designed and conducted in an construction setting. The system was comprehensively tested across non-conflict, static-conflict, and dynamic-conflict scenarios.

5.1. Experiment setting

The construction robot used in the experiment is identical to the prototype in Fig. 2a. The depth camera utilized by the robot is Obbrec Femto Bolt, providing a relative depth accuracy of 0.15 % at 1 m. The safety monitoring system was tested on an Ubuntu 20.04 platform with hardware specifications of an Intel i7–11,730 CPU and an NVIDIA RTX 3070 graphics card. Three scenarios were designed to comprehensively test the performance of the system, as shown in Fig. 9.

1) In the non-conflict scenario, the robot repeatedly moved its end-effector along a fixed route while gripping a pipe, with the worker positioning outside the robot's workspace and remaining stationary. This experiment aimed to evaluate the SMS's feasibility and its latency and accuracy in perceiving SD.



Fig. 9. Test scenarios for the experiment.

2) In the static-conflict scenario, the worker placed their hand in the robot's path to receive the material but remained stationary. This simulated a static human-robot conflict scenario, where a robot needs to stop before a collision occurs.

3) In the dynamic-conflict scenario, the worker initially stayed outside the human-robot conflict area but intervened by placing their hand in the robot's path after it had started moving. This introduced a more dynamic conflict scenario, which more closely mimicked WRC in reality, requiring the robot to quickly respond to the worker's actions and halt before a collision occurs.

Three participants were invited in the experiment, with their heights being 167 cm, 175 cm, and 181 cm, corresponding to the 50th (P50), 80th (P80), and 95th (P95) percentiles of adult static anthropometric dimensions respectively. For safety considerations, all participants were asked to wear appropriate protective equipment in the static-conflict and dynamic-conflict scenarios to prevent injuries from potential safety monitoring failures. The construction robot was supervised by an operator who ensured that the robot could be manually halted in case the SMS failed.

5.2. Result analysis

Utilizing the DT system, virtual assets and values, including spatial data related to construction robots, workers, and materials, as well as SD and PSD, were recorded and stored as datasets. The performance of the proposed SMS was analyzed using the datasets from each scenario.

5.2.1. Non-conflict scenario

In the non-conflict scenario, the primary focus of analysis was on the system's accuracy and efficiency under various WRC situations. The participants were instructed to assume various postures throughout the experiment, including standing facing forward, standing sideways, and squatting, to validate the system's effectiveness across a range of simulated WRC situations. Fig. 10 illustrates the visualization of perception and SD calculation for a worker and robot across different poses. Under stable relative postures, the perceived SD was recorded, and the actual minimum human-robot distance was measured using a tape measure. Contributions from the three participants yielded 21 sets of perception-measurement data, as plotted in Fig. 11. Special attention was given to errors where the actual measurements were less than the perceived SD, as these discrepancies could lead to overly optimistic collision assessments. The results show that the maximum SD error, where the actual measurement is less than the perceived value, is 9 cm. It means that the proposed digital twin system can accurately perceive the poses of all entities and precisely calculate SD under various postures.

To evaluate the reaction time, the construction robot executed a reciprocating horizontal motion in the X-axis direction of the base coordinate system, while the worker maintained a relatively stationary position. The robotic arm's self-position perception measurement was

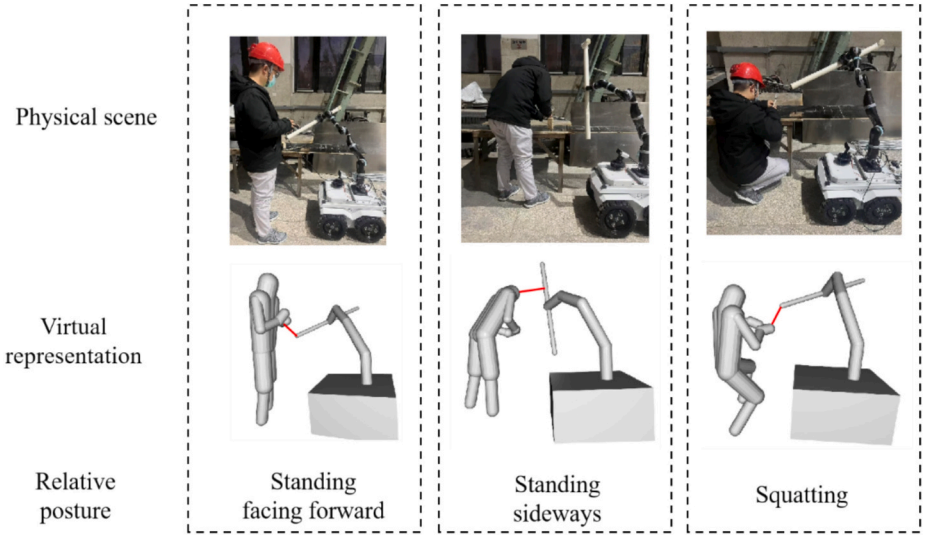


Fig. 10. Digital twin representation for different poses of worker and robot.

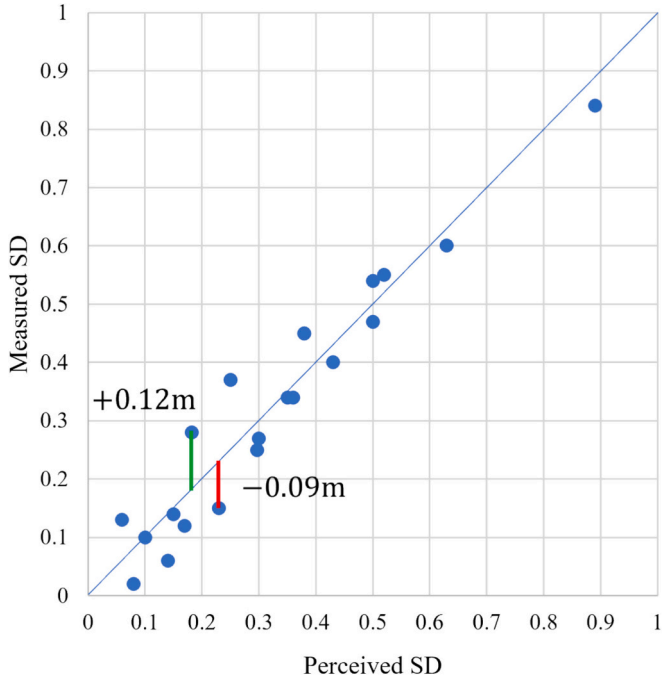


Fig. 11. Perceived errors of SD in the non-conflict scenario.

used as the time reference since it stems directly from the built-in encoders, offering the minimum computation time and closely approximating real-time synchronization to the physical world. The reaction time of the SMS could be quantified by comparing the time difference between the robot's self-position and the perceived SD reaching their extremum, respectively (Fig. 12). However, the slight swaying of workers' bodies and perception errors caused by sensors and algorithms create disturbances, which may affect the perceived SD value and the timing of SD reaching its extremum. Instead of calculating latency directly from single measurement data, the correlation coefficient between the robot's position and SD during its cyclic motion was calculated, then the average system reaction time was obtained by calculating their phase difference. The findings show that the SMS achieves a perception frame rate of approximately 9.8 frames per second (FPS) in SD perception and an average reaction time of 0.177 s. The reaction time consists of the time required for the system to perceive and integrate the poses of entities and the computation time for constructing geometric representations and calculating SD and PSD.

5.2.2. *Static-conflict scenario*

With the results of the non-conflict scenario experiment, the parameters for calculating PSD are presented in Table 3.

Each participant was asked to perform three static-conflict tests under distinct robot speeds (v_r= 0.08 m/s, 0.10 m/s, 0.12 m/s). The robot speed was defined as the linear velocity of end effector during motion execution. In all tests, the SMS reliably initiated full braking sequences across all tested velocities, achieving zero human-robot contact incidents. Fig. 13 shows the experiment result of the first participant. It can be found that the dynamic interplay among SD, PSD, and the end-effector's positional trajectory exhibits velocity-dependent scaling behavior. At different robot speeds, the average PSD values are 0.116 m, 0.125 m, and 0.145 m, respectively, which are scaled proportionally with velocity. This velocity-dependent PSD amplification aligns fundamentally with SSM principles, where higher operational velocities necessitate larger buffer zones to compensate for fixed system latency constraints.

The fluctuations in PSD primarily stem from variations in the SMS's perception of relative speed between robots and workers. Given that the workers remained relatively stationary during the robot's movement, and its end effector moved at a constant linear velocity relative to the workers, PSD remained relatively stable with limited fluctuation due to the absence of significant changes in the risk of human-robot conflict, resulting in a consistently stable PSD under the static-conflict scenario.

5.2.3. *Dynamic-conflict scenario experiment*

Each participant was asked to perform three independent tests in the

Table 3
Parameters for calculating PSD.

Parameter	Value	Explanation
T_r	0.177 s	Reaction time, including perception and computing time.
T_s	0.22 s	Stopping time, including the transmission time of the stop signal, and the braking time defined by the performance and the speed of the robot. 0.22 s is the average stopping time for the robot's maximum speed.
C	0 mm	Intrusion distance, C is zero since the worker body is guaranteed to be detected once it enters the sensor field.
$Z_d + Z_r$	90 mm	Position estimation error for the worker and the robot, equaling the maximum perception error of SD.

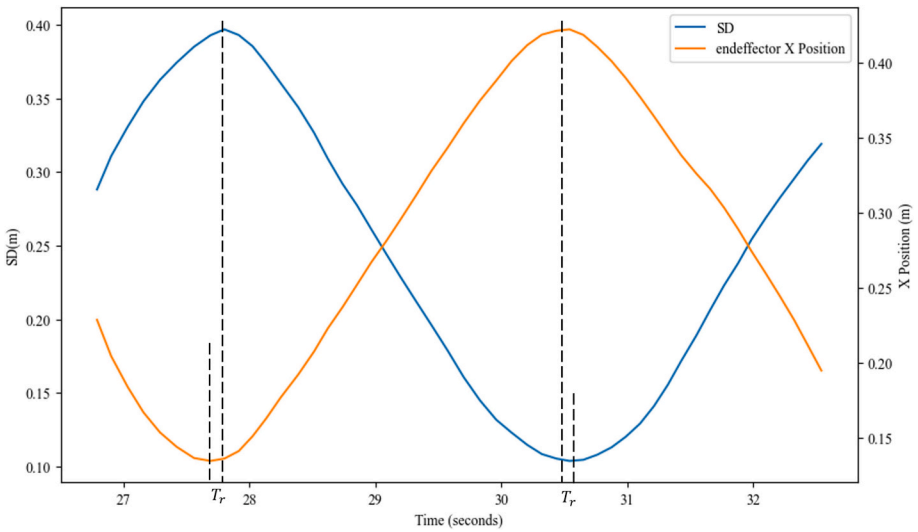


Fig. 12. Phase difference between cyclic SD and robot position over time.

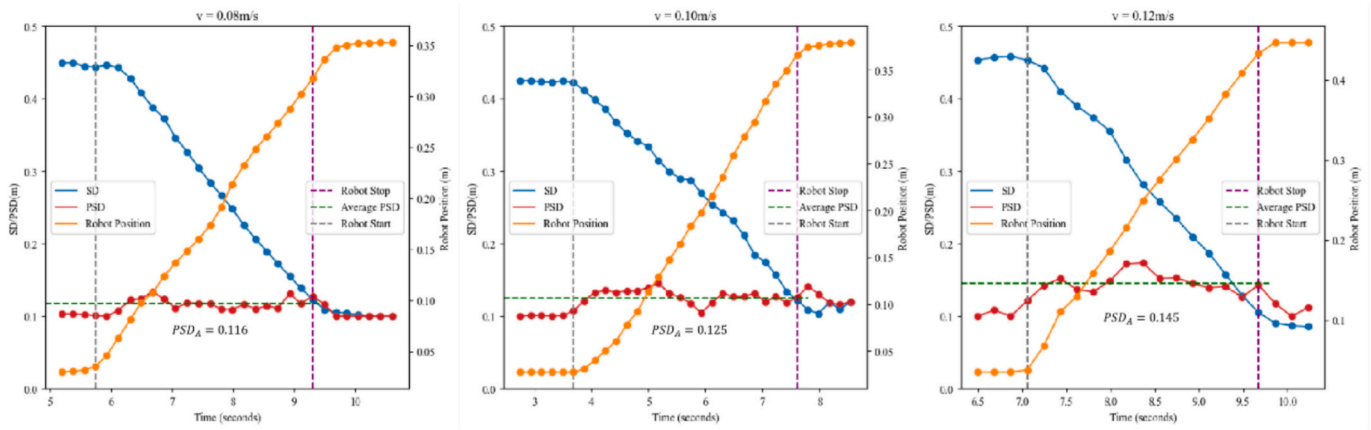


Fig. 13. SD, PSD, and robot position under different robot speeds in the static-conflict scenario.

Table 4
PSD and SD at robot stop triggering and stopping.

Experiment number	SD at t_0 (m)	PSD at t_0 (m)	SD at t_1 (m)
1	0.215	0.417	0.117
2	0.288	0.332	0.183
3	0.321	0.326	0.205
4	0.327	0.339	0.201
5	0.163	0.243	0.094
6	0.342	0.78	0.019
7	0.594	1.3	0.103
8	0.351	0.852	0.087
9	0.412	0.667	0.216

dynamic-conflict scenario. Table 4 enumerates the experiment results, encompassing SD at stop triggering (t_0), PSD at stop triggering, and SD at robot stopping (t_1). The results affirm that the proposed SMS can dynamically adjust PSD to stop the robot prior to any human-robot collision when the workers intercede. Further examination of the SMS operation is provided based on the sixth experiment, wherein SD, PSD, and robot position are illustrated in Fig. 14. Owing to the worker's dynamic intervention, SD exhibits a nonlinear and swift decrease between the fifth and sixth second. A marked escalation in PSD is observed, enabling the robot to stop at a greater SD. The SMS's performance aligns with expectations that in dynamic-conflict scenarios, the worker-to-robot position is less predictable, leading to a larger zone of potential conflict, where a greater PSD can mitigate collision risks by stopping robots at a greater SD.

5.2.4. *Comparison with other SSM methods*

To further illustrate the sophistication of the proposed SMS, quantitative and qualitative comparisons were made with other similar methods based on SSM, detailed in Table 5. Firstly, the proposed SMS monitors the entire human body and uses geometric shapes to represent the collision regions of entities, which provides a more accurate representation of entities' spatial occupancy. Therefore, it offers refined SD perception compared to methods that rely on single point or keypoints for entity representation in human-robot distance calculation. The more precise SD perception allows for closer WRC while ensuring safety, which is essential for achieving seamless WRC on site.

Secondly, the proposed SMS considers the manipulated materials by construction robots during safety monitoring, offering a more comprehensive assessment of on-site WRC risks. Meanwhile, the dynamic PSD calculation method facilitates a more flexible WRC organization, allowing workers to operate closer to robots at low speed and maintain a greater safe distance at fast speed, manifested by an automatically adjusted PSD.

Lastly, the detection speed of the SMS is 9.8 FPS, achieving finer SD perception and dynamic PSD calculation without significantly compromising computational speed, manifesting its capability for real-time safety monitoring in WRC on site.

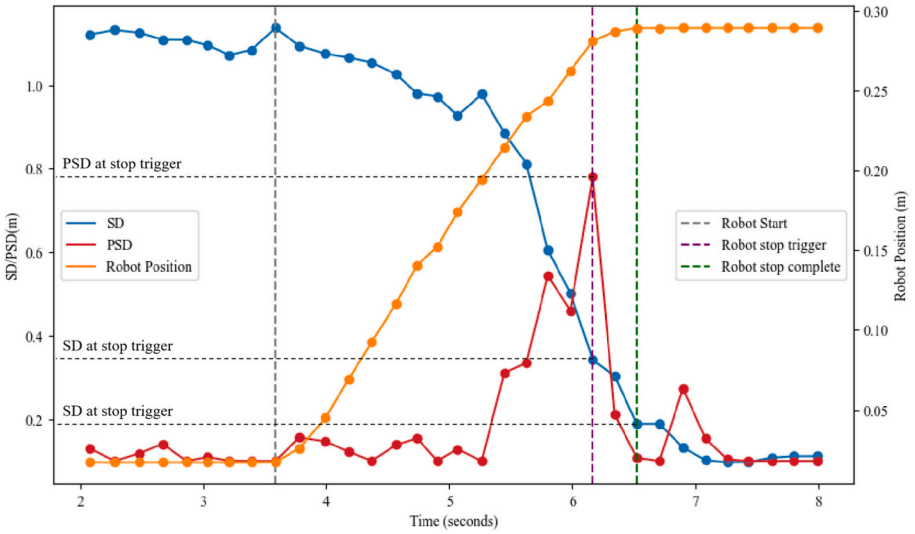


Fig. 14. SD, PSD, and robot position in the dynamic-conflict scenario.

Table 5
Comparison of the proposed SMS and other SSM-based methods.

Method	Worker body LoD	Collision representation	Manipulated object monitoring	Dynamic PSD	Monitoring rate (FPS)
Rosenstrauch et al. [29]	Whole limbs	Keypoint	No	No	Not reported
Kim et al. [33]	Hands	Keypoint	No	Yes	Not reported
Karagiannis et al. [44]	None	Single point	No	No	Not reported
Secil et al. [32]	Upper-limbs	Geometric solid	No	No	11.6
Park et al. [45]	Whole limbs	Keypoint	No	No	Not reported
The proposed SMS	Whole limbs	Geometric solid	Yes	Yes	9.8

6. Conclusion

Seamless WRC has considerable potential for enhancing productivity in construction but must be supported by adequate safety monitoring methods or systems. Although SSM is deemed suitable for WRC on site, the practicality of existing methods is limited due to that they are designed especially for industrial WRC that is less dynamic and complex. To bridge the gap, this paper introduced a digital twin-enabled safety monitoring system. The proposed SMS offers the following contributions: 1) a comprehensive perception method for WRC entities, including construction workers, robots, and materials; 2) a geometric solid-based virtual representation method for WRC entities that balances sensing accuracy and efficiency; 3) a rapid calculation method for SD, and 4) a dynamic calculation method for PSD. The above-mentioned physical entities, their virtual twins, and relevant functions are integrated within a digital twin system. Experiments have shown that the proposed SMS achieves an average monitoring rate of 9.8 FPS, an average reaction latency of 0.177 s, and an SD perceiving error of 0.09 m. This system has been validated in both static and dynamic-conflict scenarios, demonstrating its safety-guarding capabilities.

The future directions for further enhancing the proposed SMS are discussed as follows. First, this system will be validated on a broader range of construction robots beyond the mobile manipulator configuration, such as humanoid robots. Second, it will continuously benefit from more advanced perception methods for human workers and construction materials with less latency and higher accuracy. Finally, the incorporation of predictive methods for workers' and robots' motion will further enhance the performance of PSD calculation, empowering a more robust and flexible WRC on site.

CRediT authorship contribution statement

Xiao Lin: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Ziyang Guo:** Writing – review & editing, Writing – original draft, Methodology. **Xinxiang Jin:** Software, Formal analysis, Conceptualization. **Hongling Guo:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

We would like to thank the National Natural Science Foundation of China (Grant No. 52278310) for supporting this research.

Data availability

No data was used for the research described in the article.

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